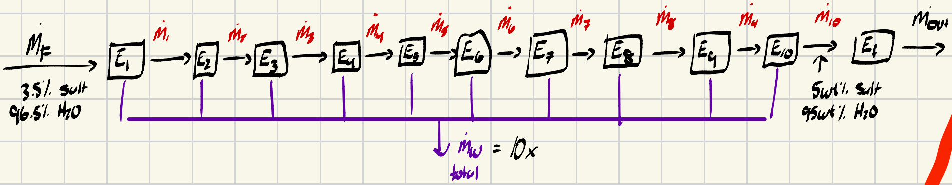
1.




1a - Since we only want to find the mass flow rate of what leaves the $10^{th}$ evaporator all we need is the $1^{st}$, $4^{th}$, and $10^{th}$ mass flow rates, and ∴ can be condensed to not show all 10 units

1b. $Acc = In - Out + \cancel{Gen} - \cancel{com}$

No rxn

Water mass balance

Water In = Water Out

$$0.965\,\dot{m}_f = 0.95\,\dot{m}_{10} + \dot{m}_w$$

Salt mass balance

Salt in = Salt out

$$0.035\,\dot{m}_f = 0.05\,\dot{m}_{10}$$

Total mass

Mass in = Mass out

$$\dot{m}_f = \dot{m}_{10} + \dot{m}_w$$

1c.

$$\dot{m}_{10} = \frac{0.035\,\dot{m}_f}{0.05}$$

$$\boxed{\dot{m}_{10} = 0.7\,\dot{m}_f}$$

$$0.965\dot{m}_f = 0.95\dot{m}_{10} + \dot{m}_w$$

$$0.965\dot{m}_f = 0.95(0.7\,\dot{m}_f) + \dot{m}_w$$

$$0.965\dot{m}_f = 0.665\dot{m}_f + \dot{m}_w$$

$$\dot{m}_w = 0.965\dot{m}_f - 0.665\dot{m}_f$$

$$\dot{m}_w = 0.3\,\dot{m}_f$$

We need a basis

Let $\dot{m}_f = 1000$ kg/hr

$$\dot{m}_{10} = 0.7\left(1000\ \tfrac{kg}{hr}\right) = 700\ \tfrac{kg}{hr}$$

$$\dot{m}_w = 0.3\left(1000\ \tfrac{kg}{hr}\right) = 300\ \tfrac{kg}{hr}$$

Mass fraction - $\dot{m}_f = (4 \cdot 0.1)\dot{m}_w + \dot{m}_4$

fractional yield - $\dfrac{0.3\dot{m}_f}{0.965\dot{m}_f}$ $= 0.311$

Recovered

feed

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$$1000 = 4 \cdot 0.1\,\dot{m}_w + \dot{m}_4$$

$$1000 = 0.4\left(300\ \tfrac{kg}{hr}\right) + \dot{m}_4$$

$$\dot{m}_4 = 1000 - (0.4 \cdot 300)$$

$$= 880\ \tfrac{kg}{hr}$$

$$0.035 \cdot \dot{m}_f = w_{4s} \cdot \dot{m}_4$$

$$0.035\left(1000\ \tfrac{kg}{hr}\right) = w_{4s}\left(880\ \tfrac{kg}{hr}\right)$$

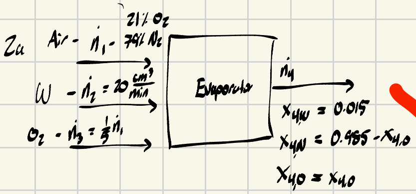
$$w_{4s} = \frac{0.035 \cdot 1000}{880}$$

$$= 0.03977\ \frac{kg\ salt}{kg\ solution}$$

$$w_n = \frac{0.035 \cdot 1000}{1 - 0.1x}$$

2

2a




2b

DoF analysis

4a - # of components in all streams

4b - # of rxn's / accumulations

5a - specified flows

5b - specified comp

5c - System specifications

5d - species at system boundary

4a - 7 - 4 in, 3 out

4b - 0

7

5a - 1 - $20 \frac{cm^3}{min}$

5b - 3 - $\dot{n}_1, \dot{n}_3, \dot{n}_4$

5c - 0

5d - 3 - $x_{4W}, x_{4N}, x_{4,O}$

7

DoF = 7 - 7 = 0

∴ well posed

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2c. $M_{H_2O} = 18.015 g/mol$

$\rho_{H_2O} = 1 g/cm^3$

$20 \frac{cm^3}{min} \times 1 \frac{g}{cm^3} \times \frac{1}{18.015} \frac{mol}{g} = 1.11 \frac{mol}{min} = \dot{n}_2$

Total mass balance

$\dot{n}_1 + \dot{n}_2 + \dot{n}_3 = \dot{n}_4$

$\dot{n}_3 = \frac{1}{5}\dot{n}_1$

$\dot{n}_2 = x_{4,W} \cdot \dot{n}_4$

$1.11 \frac{mol}{min} = 0.015 \cdot \dot{n}_4$

$\dot{n}_4 = \frac{1.11}{0.015} = 74 \frac{mol}{min}$

$\dot{n}_1 + \dot{n}_2 + \frac{1}{5}\dot{n}_1 = 74 \, mol/min$

$\dot{n}_1 + 1.11 \frac{mol}{min} + \frac{1}{5}\dot{n}_1 = 74 \frac{mol}{min}$

$\dot{n}_1 + \frac{1}{5}\dot{n}_1 = 74 - 1.11$

$\frac{5}{5}\dot{n}_1 + \frac{1}{5}\dot{n}_1 = 72.89 \frac{mol}{min}$

$\frac{6}{5}\dot{n}_1 = 72.89$

$\dot{n}_1 = \frac{72.89 \cdot 5}{6}$

$\dot{n}_1 = 60.74 \frac{mol}{min}$

$\dot{n}_3 = \frac{1}{5}\dot{n}_1$

$= \frac{1}{5} 60.74$

$\dot{n}_3 = 12.15 \frac{mol}{min}$

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2.66 $TiO_2$ - Washer to handle 10,000 lb/hr of raw pigment w/ 40% $TiO_2$, 20% salt, 40% water

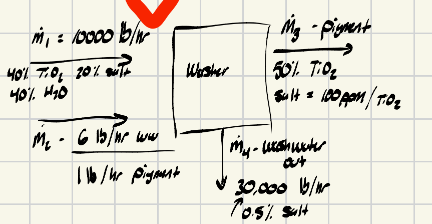
- Mixed w/ washwater @ 6 lb washwater / 1 lb of pigment
- 100% $TiO_2$ is recovered in washed pigment
- Washed pigment pdt - 50% $TiO_2$ w/ salt < $\frac{100 \text{ parts salt}}{1 \cdot 10^6 \text{ parts pigment}}$
- Washwater is dumped in river

W/ limit of 30,000 lb/hr containing 0.5% salt

1. Draw flow diagram
2. Do DoF, is it specified?
   - If not what needs to change to fix problem

25/25

1.




$\dot{m}_1 = 10000$ lb/hr - Total

$\dot{m}_{1,TiO_2} = 4000$ lb/hr $TiO_2$

$\dot{m}_{1,salt} = 2000$ lb/hr Salt

$\dot{m}_{1,H_2O} = 4000$ lb/hr $H_2O$

Washwater $= 10,000 \frac{lb}{hr}$ Pigment $\times 6 \frac{lb}{hr}$ ww $= 60,000 \frac{lb}{hr}$ ww

Washed pigment: 4000 lb/hr (In) $= 0.5 \cdot \dot{m}_{3,TiO_2}$ (Out)

$\dot{m}_{3,TiO_2} = \frac{4000 \text{ lb/hr}}{0.5}$

$= 8000$ lb/hr $TiO_2$

DoF

4a - 9

4b - 0

5a - 2

5b - ~~5~~ $^*(N-1) = 4$

5c - 2

5d - 3

DoF = 9 - 11 = -2

∴ The problem is overspecified

total Salt mass balance

$Salt_{\dot{m}_1} = Salt_{\dot{m}_3} + Salt_{\dot{m}_4}$

$2000 \frac{lb}{hr} = 0.4 \frac{lb}{hr} + \dot{m}_4$

$\dot{m}_4 = 1999.6 \frac{lb}{hr}$

Salt balance

$4000 \times \frac{100}{1 \cdot 10^6} = 0.4 \frac{lb}{hr}$ salt ($\dot{m}_3$)

$Salt_{max}$ in $\dot{m}_4$

$30,000 \cdot 0.05 = 1500$

- There is too much salt in $\dot{m}_4$, splitter would have to recycle flow rate

ww + raw pnt - washed pnt

(60,000 + 10000) (In) - 8000 (Out) $= 62000$ lb/hr $= \dot{m}_4$

optimum ww $\dot{m}_2$ rate % $= \frac{1999.6 \frac{lb}{hr}}{62,000 \frac{lb}{hr}} \cdot 100 = 3.23\%$

2.69

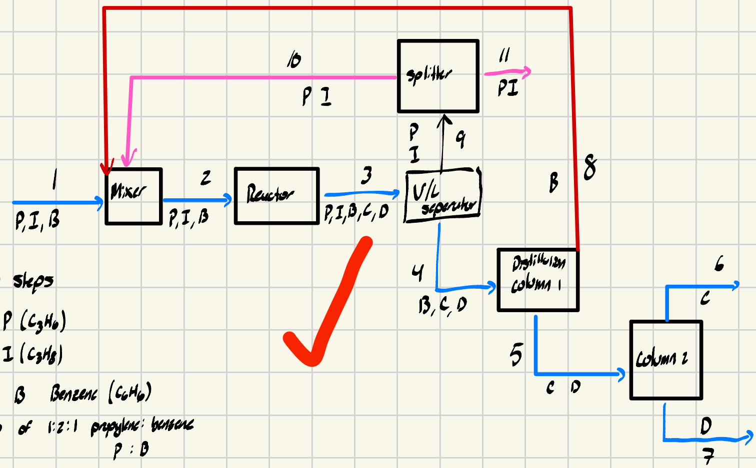



Diagram steps

1. 95 mol % P ($C_3H_6$)

   5 mol % I ($C_3H_8$)

mixed w/ B Benzene ($C_6H_6$)

w/ mol ratio of 1.2:1 propylene : benzene
P : B

2. Fed in recycled streams

   Fed into reactor

   $\dot{n}_2$ = 10% I

3. 2 rxns

   1- Cumene C - ($C_9H_{12}$)

   2- diisopropylbenzene D - ($C_{12}H_{18}$)

   1rxn - $C_3H_6 + C_6H_6 = C_9H_{12}$

   2rxn - $C_3H_6 + C_9H_{12} \to C_{12}H_{18}$

* I is inert
* 80% P + 72% B converted to C + D

Cooled, sent to separator

9 - P, I in vapor stream

4 - B, C, D in liquid stream

8 - part of stream is purged

5 - seperated again

C's production rate - 25 gmol/s

Important info to consider

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* Do a DoF analysis

- Write every variable + Eq to calculate flow rates

4a

| | |
|---|---|
| 3 | $S_1$ |
| 3 | $S_2$ |
| 5 | $S_3$ |
| 3 | $S_4$ |
| 2 | $S_5$ |
| 1 | $S_6$ |
| 1 | $S_7$ |
| 1 | $S_8$ |
| 2 | $S_9$ |
| 2 | $S_{10}$ |
| 2 | $S_{11}$ |
| 25 | |

4b - 2 rxns

27

5a - 1

5b - 4

5c - 2

5d - 3 Mixer
5 Reactor
5 V/L
2 S
3 $DC_1$
2 $DC_2$

27

DoF = 27 - 27 = 0 ∴ Well posed

Material balances for seperate components

Mixer
$P_2 = P_1 + P_{10}$
$I_2 = I_1 + I_{10}$
$B_2 = B_1 + B_8$

Reactor
$P_3 = P_2 - P_1 - P_2$
$I_3 = I_2$
$B_3 = B_2 - B_1$
$C_3 = C_1 - C_2$
$D_3 = D_2$

V/L
$P_9 = P_3$
$B_4 = B_3$
$C_4 = C_3$
$D_4 = D_3$
$I_9 = I_3$

Splitter
$P_9 = P_{10} + P_{11}$
$I_9 = I_{10} + I_{11}$

$DC_1$
$B_4 = B_8$
$C_4 = C_5$
$D_4 = D_5$

$DC_2$
$C_5 = C_6$
$D_5 = D_7$